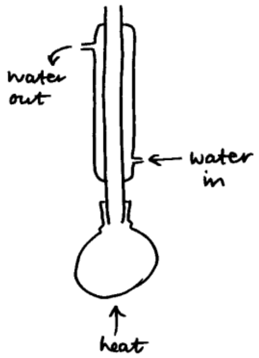
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		appropriate scales (1) all points plotted correctly (1) (tolerance ±1 square)		2		2	2	2
		(ii)		straight line extrapolating back to y-axis (1) ΔT = 44 – 17 = 27 (1) where vertical scale covers 0-40 extrapolation back to y-axis is not possible so mark as follows line of best fit drawn (1) ΔT = 41 – 17 = 24 (1)			2	2	2	1
		(iii)		mass of solution = 380g (1) $q = -mc\Delta T$ = 380 × 4.18 × 27 = – 42 886.8 J (1) n($CH_3COONa.3H_2O$) = $\frac{320}{136.09}$ = 2.35 mol (1) $\Delta H = \frac{-42.9}{2.35}$= –18.26 kJ mol^{-1} (1) ECF possible ΔH = –16.22 using ΔT = 24 from part (ii)		4		4	2	1
		(iv)		heat is lost to the surroundings / sodium ethanoate does not fully crystallise			1	1		1
	(b)			award (1) for each reactant $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$ award (2) for correct equation using Na_2CO_3 $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$		2		2		

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(c)	(i)		ethanol / ethanal and acidified potassium dichromate (1) colour change from orange to green (1) OR ethanol / ethanal and acidified potassium manganate(VII) (1) colour change from purple/light pink to colourless (1)		2		2		2
		(ii)		 condenser with water in at bottom and out at top (1) condenser attached upright to top of reaction vessel (1) source of heat (with upright condenser drawn) (1) if distillation apparatus drawn – (1) max for water in / water out			3	3		3
				Question 10 total	**0**	**10**	**6**	**16**	**8**	**9**